



Great Lakes St Lawrence Seaway Development Corporation (GLS)

Performance Work Statement Computerized Maintenance Management System (CMMS) Implementation

1.0 Background

The Great Lakes St. Lawrence Seaway Development Corporation (GLS) is a federal agency within the U.S. Department of Transportation responsible for the operation and maintenance of the U.S. portion of the St. Lawrence Seaway between Lake Ontario and Montreal, Quebec. GLS operates and maintains two locks, supports vessel traffic operations, and manages associated marine navigation, maintenance, and administrative functions critical to safe and efficient commercial navigation.

To support this mission, GLS relies on accurate and timely management of maintenance activities, work orders, inventory control, and preventive maintenance (PM) scheduling. These functions are currently supported by a legacy Computerized Maintenance Management System (CMMS) that no longer fully meets operational, security, reporting, and integration requirements.

GLS requires a modern, cloud-based, FedRAMP-authorized CMMS to improve operational efficiency, strengthen cybersecurity posture, support regulatory and audit requirements, and provide real-time visibility into maintenance and inventory operations for management, technicians, and administrative personnel.

2.0 Scope of Work

The Contractor shall provide a Commercial Off-The-Shelf (COTS) Computerized Maintenance Management System (CMMS) solution. The Contractor shall be responsible for full implementation, configuration, data migration, training, testing support, and post-implementation operational support.

The system shall be delivered as a cloud-based solution hosted in a FedRAMP-authorized environment and shall be configured (not custom source-code modified) to meet GLS functional requirements.

The Contractor shall ensure the system is fully operational, meeting all functional performance, security, and data migration requirements defined in this PWS, and accepted in accordance with Section 8.0.

3.0 System Requirements

3.1 General System Requirements

The CMMS shall:

- Be a cloud-based solution, FedRAMP-authorized with an impact level of Moderate or higher.
- Be accessible via modern web browsers without degradation of functionality.
- Be fully functional on desktop and mobile devices (phones and tablets).
- Provide role-based access control for all users.
- Maintain a user-friendly interface with built-in help functions, job aids, and system prompts.
- Be Section 508 compliant in accordance with applicable accessibility standards.
- Maintain a full audit trail for all system transactions including work orders, inventory, and preventive maintenance activities.

3.2 Work Order and Preventative Maintenance Requirements

The CMMS shall:

- Support creation, assignment, tracking, completion, and archival of work orders (support computer and mobile devices).
- Support preventive maintenance scheduling including time-based, usage-based, and event-based triggers.
- Provide scheduling tools that allow assignment of personnel.
- Provide asset hierarchy management with at least three levels (e.g., location, area, asset).
- Allow attachments including documents, photos, and supporting files within work orders.
- Provide searchable historical records for all work orders and maintenance activities.
- The system shall support configurable approval workflows for assigning and tracking work orders.

3.3 Inventory and Asset Management Requirements

The CMMS shall:

- Support full inventory lifecycle management including tracking by location, expiration dates, rejected issue transactions, mismatch issues, obsolete inventory, assemblies/component parts, and cycle counts.
- Support inventory valuation methods including FIFO, actual cost, receipt costing, and last received cost options.
- Provide asset hierarchy management for specific inventory items (spare parts) to link directly to assets within the asset tree.
- Maintain full transaction history including auditability of all inventory changes.
- Support supplier tracking metrics including delivery performance and quality indicators: on-time delivery (OTD), lead time variability, fill rates, rejection rates, and invoice accuracy.
- The system shall support configurable approval workflows for inventory transactions, including supervisor approval for designated actions.
- Allow attachments including documents, photos, and supporting files within inventory items.

3.4 Reporting and Analytics Requirements

The CMMS shall:

- Provide configurable reporting tools accessible to users based on role permissions.
- Allow export of data in common formats including CSV, PDF, XML, and Excel-compatible formats.
- The system shall allow users to customize reports, including the ability to include or exclude specific data fields.
- Provide role-based graphical dashboards and visual reporting capabilities for operational and management use.

- Support queries based on time, location, asset type, technician, and work order status.
- Provide data in number and percent (%) formats, organizing work order data into statistical categories including trends, means, deviations, and cumulative data.
- Part-level breakdowns, location tracking, reorder needs, job due reports, parts on hand, inventory valuation by class, inventory value reconciliation, and purchasing/receiving transaction reports.
- The system shall provide performance metrics including, at a minimum, Mean Time to Repair (MTTR) and work order completion metrics.

3.5 Integration Requirements

The CMMS shall:

- Provide integration capability with external systems via standard APIs or standard data exchange methods.
- Allow import/export of data without requiring modification to Government systems.
- Support structured data feeds including CSV and XML formats.

3.6 Security and Compliance Requirements

The CMMS shall:

- Be hosted in a FedRAMP-authorized cloud environment, and an impact level of Moderate or higher. These designations must be current at the time of award and maintained throughout the period of performance.
- Maintain system logs for security monitoring and audit purposes.

3.7 User Roles and Access Management

The CMMS shall provide role-based access control (RBAC) supporting configurable user roles and permissions. The system shall ensure users are granted the minimum level of access necessary to perform assigned duties.

The system shall, at a minimum, support the following user roles:

Level 1 – Requestor Role

- Submit maintenance requests
- View status of submitted work orders
- Read-only access to assigned requests

Level 2 – Technician Role

- Receive, update, and close assigned work orders
- Record labor, materials, and completion data
- Access work order history relevant to assigned duties

Level 2 – Inventory Technician Role

- Manage inventory transactions including receiving, issuing, transfers, and cycle counts
- Access inventory valuation and reporting functions (FIFO / Actual Costing)
- Support asset and parts tracking functions

Level 3 – Management Role

- View and analyze work order and inventory performance metrics
- Manage work orders across departments
- Approve inventory transactions as required
- Access reporting dashboards and operational analytics

Level 4 – Administrator Role

- Create and manage user accounts and role assignments
- Configure system settings, workflows, and asset hierarchies
- Manage system integrations, backups, and security configurations

User Capacity Requirements

The system shall support, at a minimum:

- Minimum of 50 Level 1 users (Requestors)
- Minimum of 50 Level 2 users (Technicians/Inventory Technicians)
- Minimum of 10 Level 3 users (Management)
- Minimum of 5 Level 4 users (Administrators)

Licensing shall be structured to support either named, concurrent, or hybrid models, as proposed by the Contractor.

3.8 Historical Data and Records Management

The CMMS shall provide the ability for at least 10 years of historical data retention, searchability, and retrieval capabilities to support operational, audit, and reporting requirements.

3.8.1 Work Order and Preventive Maintenance History

The system shall store and allow queries of historical work order and preventive maintenance (PM) data using multiple search parameters, including but not limited to:

- Work order type
- Location
- Time period
- Technician
- Status

3.8.2 Asset History

The system shall provide a searchable, comprehensive history for each asset, including but not limited to:

- Full lifecycle maintenance history
- Repair and service records
- Asset movement and location changes

- Associated labor and material costs
- Status changes over time

3.8.3 Inventory History

The system shall provide a complete transaction history for all inventory items, including but not limited to:

- Receipt transactions
- Issue transactions
- Transfers and adjustments
- Cycle counting activity
- Cost changes and valuation adjustments

The system shall maintain a full audit trail of all inventory-related changes. The system shall prevent or log any “silent” or untracked changes to inventory data, including cost or quantity modifications.

4.0 Data Migration Requirements

The Contractor shall be responsible for full migration of legacy CMMS data into the new system.

4.1 Data Scope

At minimum, migration shall include:

- Active and historical work orders
- Preventive maintenance records
- Inventory records and transactions
- Asset registries and hierarchies
- Personnel records relevant to system operations
- Attachments and supporting documentation

Migration is limited to a maximum of five (5) years of historical data and attachments.

4.2 Government-Furnished Data

GLS will provide legacy data extracts in structured formats (e.g., CSV or XML) and may include PDF attachments associated with legacy data.

4.3 Contractor responsibilities

The Contractor shall:

- Develop and execute a data migration plan.
- Validate data accuracy and completeness post-migration.
- Support reconciliation between legacy and new system data.

4.4 Acceptance

Migration shall not be considered complete until GLS validates that migrated data is accurate, accessible, and functional with the new system.

4.5 Data Ownership

All data created, stored, or processed within the CMMS shall remain the property of the Government. The Contractor shall provide the ability to export all system data in a structured, non-proprietary format (e.g., CSV, XML) with associated PDF or other types of attachments, upon request and at contract completion.

The Contractor shall support transition of data to a successor system without loss of data integrity or accessibility.

4.6 Transition Support

The Contractor shall provide transition assistance at the end of the contract period to support migration to a successor system or Government-operated solution. This includes data export, system documentation, and technical support for a period of up to 60 days after contract completion.

5.0 Implementation and Configuration

The Contractor shall:

- Configure the system using out-of-the-box capabilities.
- Develop workflows, dashboards, and reporting structures.
- Support system testing and go-live activities.
- Provide a cutover plan for transition from the legacy system.

5.1 Project Management Plan (PMP)

The Contractor shall develop and submit a Project Management Plan (PMP) within 15 calendar days of contract award and prior to commencement of implementation activities.

The PMP shall describe the Contractor's approach to planning, executing, and controlling Base Year implementation activities, including at a minimum:

- Project organization and roles
- Implementation methodology
- Work breakdown structure (WBS)
- Schedule and milestone plan
- Resource management approach
- Risk management approach
- Quality assurance approach
- Communication plan
- Training Plan

The PMP shall be subject to Government review and approval by the COR.

5.2 Project Reporting Requirements

During the Base Year implementation period, the Contractor shall provide periodic status reporting to the Government.

Reports shall include, at a minimum:

- Summary of completed work and milestones
- Status of ongoing tasks and deliverables
- Schedule variances and recovery actions
- Risks and issues with mitigation plans
- Personnel changes affecting performance

Reporting frequency shall be:

- Weekly during active implementation
- Monthly executive summary report

5.3 Project Management Reviews

The Contractor shall participate in monthly Project Management Reviews (PMRs) with the COR and designated Government stakeholders.

PMRs shall address:

- Project status and milestone progress
- Risks and issue resolution status
- Schedule performance
- Resource adequacy
- Upcoming activities and dependencies

PMRs may be conducted via teleconference or in-person at Government discretion.

6.0 Training and Documentation

The Contractor shall provide role-based training and documentation.

The Contractor shall provide a comprehensive training program to ensure all GLS personnel are proficient prior to the full system "Go-Live" date.

Training shall include, but not be limited to:

- **System Administrator Training:** A minimum of 16 hours of in-person deep-dive technical training for up to five (5) designated GLS system administrators. This shall cover system configuration, user permission management, data migration verification, and backend integration maintenance.
- **Key Role Individual Training:** Specialized training (minimum 4 hours per role) for key functional leads, including Schedulers, Inventory Technicians, and Trades Technicians, focusing on role-specific workflows and reporting.
- **Train-the-Trainer (TTT):** A dedicated TTT event for up to five (5) GLS "super users" (minimum 16 hours) to provide them with the knowledge and materials necessary for first-line internal support.
- **On-Site Group Training:** Five (5) business days of on-site training at the GLS facilities in Massena,

NY. This shall include distinct group sessions each day tailored to Requestors, Technicians, and Management.

Option Years 1–4: Sustaining Training and Updates

- **Annual Refresher Training:** 16 hours of remote, instructor-led training annually for designated user roles.
- **New Release Training:** Within thirty (30) days of any significant system update, the Contractor shall provide a 2-hour webinar and updated documentation.
- **On-Demand Training Access:** Maintenance of a digital "Knowledge Base" training aids accessible to all licensed users 24/7, containing up-to-date documentation and self-paced modules.

On-site training shall be conducted at a GLS facility in Massena, NY, located at either:

- 180 Andrews Street, Massena, NY, or
- 251 Fregoe Road, Massena, NY

The COR will designate the final location.

The Contractor shall provide:

- User manuals – digital delivery
- Quick reference guides
- Training materials and recorded content of common tasks (e.g., creating a work order, performing a cycle count).

7.0 Service Level Requirements

7.1 System Availability

The system shall maintain 99.9% uptime, excluding scheduled maintenance windows. Uptime is defined as the ability for users to log in, query data, and perform core functions (Work Orders and Inventory).

Planned downtime must be scheduled at least 72 hours in advance and occur outside GLS regular operating hours (6:00 AM – 4:00 PM EST, Monday through Friday).

7.2 Backup and Recovery

- Backups shall be performed at least every 24 hours.
- The system shall support restoration of data within defined recovery objectives.

The Contractor shall meet the following recovery objectives:

- Recovery Time Objective (RTO): System restoration within 4 hours of a system failure.
- Recovery Point Objective (RPO): Maximum data loss shall not exceed 24 hours.

7.3 Annual System Health Check

In the final quarter of each performance year, the Contractor shall conduct a remote review of system performance, storage utilization, and configuration optimization.

7.4 Help Desk Support

The Contractor shall provide 24/7/365 support with defined escalation procedures for critical incidents.

The Contractor shall provide a multi-tier technical support system available 24 hours per day, 7 days per week, 365 days per year to support system availability and incident resolution.

Support shall be provided in accordance with the following minimum service levels:

Priority Level	Description	Initial Response	Resolution Goal
Critical	System-wide outage or total loss of a core function	1 hour	4 hours
High	Significant functional degradation (e.g., scanners not syncing)	2 hours	8 hours
Medium	Minor functional issues or “how-to” questions	4 hours (business) / 12 hours (other)	2 business days
Low	Cosmetic issues, non-functional defects, or documentation errors that do not impact system operation	1 business day	Resolution in next scheduled release or within mutually agreed time frame, not to exceed 30 calendar days

The Contractor can propose similar but different response levels and timing, subject to approval by the designated COR. The Contractor shall ensure sufficient staffing and resources are available to meet the above service levels across all support hours. This includes maintaining adequate coverage for simultaneous incidents and peak usage periods without degradation of response or resolution performance.

The Contractor shall provide monthly SLA performance reports summarizing uptime, incident response metrics, resolution times, and any SLA breaches.

8.0 Acceptance Criteria

System acceptance shall be based on successful completion of:

- Functional testing
- User acceptance testing (UAT)
- Data migration validation
- Security and compliance verification

The system shall not be considered accepted until all critical deficiencies are resolved and all acceptance criteria are met to the satisfaction of the Government.

9.0 Optional Capabilities:

The following are optional capabilities and may be proposed by the Contractor:

- GIS integration: Compatible with Geographic Information Systems (GIS) for vehicle tracking or buoy positioning displays.
- Barcode scanning and mobile peripheral devices: If proposed, barcode scanners should be capable of operating without continuous Wi-Fi connectivity and support extended operational range.

10.0 Kick-Off Meeting

The Contractor shall attend a kick-off meeting to initiate the contract and establish a shared understanding of project objectives, requirements and expectations. The kick-off meeting serves as an important platform for effective communication and collaboration among stakeholders. The kick-off meeting shall be within 5 days of contract award, unless otherwise directed by the Contracting Officer (CO). The following individuals shall attend the kick-off meeting:

A. Contractor Representatives
Key Personnel

B. Government Representatives
Contracting Officer's Representative
Contracting Officer

The Contractor shall document the proceedings of the kick-off meeting, including action items, decisions, and any key points discussed, and provide meetings minutes to all attendees for review and recordkeeping purposes

11.0 Deliverables

The Contractor shall provide the following deliverables. All deliverables are subject to Government review and acceptance.

The Government shall review deliverables within 15 business days of submission and provide acceptance, conditional acceptance, or rejection with comments. The Contractor shall address deficiencies and resubmit within timelines agreed upon with the COR.

All deliverables shall be submitted in accordance with the timelines established in the approved Project Management Plan (PMP), unless otherwise specified herein.

11.1 Project Management Deliverables

- Project Management Plan
- Integrated Master Schedule
- Risk Management Plan
- Cutover and Implementation Plan

11.2 System Deliverables

- System Configuration Documentation

- Workflow and Dashboard Configuration Documentation
- Integration Design Documentation
- User Roles and Permissions Configuration Documentation
- Audit Trail Configuration and Validation Documentation
- Security and Compliance Documentation (including FedRAMP authorization evidence and system security overview)
- Integration Testing and Validation Report
- Reporting and Dashboard Configuration Documentation
- Sample Standard Reports and Dashboard Outputs

11.3 Data Migration Deliverables

- Data Migration Plan within 30 days of award
- Data Mapping Documentation
- Migration Execution Report
- Data Reconciliation and Validation Report

11.4 Testing Deliverables

- System Test Plan prior to system testing activities
- UAT Plan and Results
- Defect Tracking and Resolution Log

11.5 Training Deliverables

- Training Plan
- Training Materials (role-based)
- User Guides and Quick Reference Materials
- Train-the-Trainer Materials
- Training Completion Report (including attendance records and summary of training delivered)

11.6 Operational Deliverables

- Go-Live Readiness and Approval Report prior to production deployment
- System Administration Guide
- Help Desk Procedures
- Backup and Recovery Procedures Documentation

11.7 Post Implementation Deliverables

- Monthly SLA Performance Reports, including:
 - System uptime metrics
 - Incident response and resolution times by priority level
 - Number and type of support tickets
 - SLA compliance and any breaches

- Corrective actions taken
- Annual System Health Reports

12. Period of Performance

The Period of Performance shall consist of a 12-month Base Year and four (4) 12-month option years.

12.1 Base Year

The Base Year shall include system implementation, configuration, data migration, testing, training, and go-live support.

12.2 Option Years 1–4

Option years shall include system hosting, maintenance, help desk support, updates, and training refreshers.

13.0 Contracting Officer's Representative (COR)

The Contracting Officer's Representative (COR) monitors all technical aspects of the contract and assists in contract administration. The COR ensures that the Contractor performs the technical requirements of the contract, performs inspections necessary in connection with contract performance, maintains written and oral communications with the Contractor concerning technical aspects of the contract, and monitors Contractor performance.